

Evripidis Gkanias

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Gender: Male

Research Experience

Jun 2014	<u>Research Assistant: Machine Learning</u>	
Aug 2015	<i>Visual Computing Lab, Centre for Research and Technology–Hellas (CERTH), Greece</i>	
Sep 2016	<u>Research Assistant: Bio-Robotics</u>	
Feb 2017	<i>School of Informatics, The University of Edinburgh, United Kingdom</i>	
Mar 2017	<u>Research Associate: Computational Modelling</u>	
Aug 2018	<i>School of Computer Science, The University of Sheffield, United Kingdom</i>	
Jun 2022	<u>Postdoctoral Researcher: Computational and Neuromorphic Modelling</u>	
Dec 2024	<i>School of Informatics, The University of Edinburgh, United Kingdom</i>	
Jan 2025	<u>Postdoctoral Researcher: Computational Sensory Biology</u>	
Dec 2028 (expected)	<i>Lund Vision Group, Department of Biology, Lunds Universitet, Sweden</i> • Funding: Carl Tryggers Foundation (CTS 24: 3569). • Advisor: Prof Marie Dacke. • Extract 3D meshes of beetle eyes from μCT scans to analyse their visual resolution. • Study the impact of environmental and self-motion noise on heading calculations in dung beetles. • Prototype a robust, adaptive celestial compass sensor, incorporating behaviour and anatomy.	

Education

Sep 2008	<u>BSc in Computer Science, with specialisation in Information Systems</u>	
Jul 2013	<i>School of Informatics, Aristotle University of Thessaloniki, Greece</i> First-class honours (85%) <u>Thesis</u> : Deep learning algorithms for multi-label data.	
Sep 2015	<u>MSc in Artificial Intelligence, with specialisation in Machine Learning</u>	
Aug 2016	<i>School of Informatics, The University of Edinburgh, United Kingdom</i> Distinction (75%) <u>Thesis</u> : Data-driven adaptation of the evasion behaviour in fiddler crabs ⁸ .	
Sep 2018	<u>PhD in Bio-inspired Robotics and Autonomous Systems</u>	
Jul 2023	<i>School of Informatics, The University of Edinburgh, United Kingdom</i> <u>Thesis</u> : Insect neuroethology of reinforcement learning ³ .	

Prices & Awards

2015	The University of Edinburgh UK/EU Master's Scholarship Award.	£ 10 000
2016	Microsoft and Skype Hackathon Challenge Winner: Artificial Intelligence Driven Bots.	€ 500
2018	EPSRC Robotics and Autonomous Systems CDT Scholarship Award.	£ 76 800
2019	Percy Sladen Memorial Fund Grant: "Featureless navigation: what can we learn from ants?"	£ 1 500
2025	University of Washington—Postdoc Travel Award: Bio-inspired Sensing, Computing, and Control with International Teams (BISCCITs) Workshop, in Boulder, Colorado.	\$ 3 500
2026	The Royal Physiographic Society of Lund (Kungliga Fysiografiska Sällskapet i Lund)—Postdoctoral Research Fellow Grant: "Capturing the view of dung beetles in naturally disturbed skies".	SEK 19,644

List of publications

Journal publications

1. **Gkanias, E.**, Risse, B., Mangan, M., Webb, B. 2019. *From skylight input to behavioural output: a computational model of the insect polarised light compass*. PLoS Computational Biology, vol. 15 (7), p. e1007123.
DOI: [10.1371/journal.pcbi.1007123](https://doi.org/10.1371/journal.pcbi.1007123)
2. Schwarz, S., Clement, L., **Gkanias, E.**, Wystrach, A. 2020. *How do backward-walking ants (Cataglyphis velox) cope with navigational uncertainty?* Animal Behaviour, vol. 164, p. 133-142.
DOI: [10.1016/j.anbehav.2020.04.006](https://doi.org/10.1016/j.anbehav.2020.04.006)
3. **Gkanias, E.**, McCurdy, L. Y., Nitabach, M. N., Webb, B. 2022. *An incentive circuit for memory dynamics in the mushroom body of Drosophila melanogaster*. Elife, vol. 11, p. e75611.
DOI: [10.7554/eLife.75611](https://doi.org/10.7554/eLife.75611)
4. **Gkanias, E.**, Mitchell, R., Stankiewicz, J., Khan, S., Mitra, S., Webb, B. 2023. *Celestial compass sensor mimics the insect eye for navigation under cloudy and occluded skies*. Communications Engineering, vol. 2, p. 82.
DOI: [10.1038/s44172-023-00132-w](https://doi.org/10.1038/s44172-023-00132-w)
 - **Featured in Editors' Choice 2023**
5. Pabst, K., **Gkanias, E.**, Webb, B., Homberg, U., Endres, D. 2024. *Angular velocity integration in a locust heading circuit*. PLoS Computational Biology, vol. 20 (12), p. e1012155.
DOI: [10.1371/journal.pcbi.1012155](https://doi.org/10.1371/journal.pcbi.1012155)
6. Alcer, D., Zaiats, N., Jensen, T. K., Philip, A. M., **Gkanias, E.**, Ceberg, N., Das, A., Flodgren, V., Heinze, S., Borgström, M., Webb, B., Laursen, B. W., Mikkelsen, A. 2025. *Integrating molecular photoswitch memory with nanoscale optoelectronics for neuromorphic computing*. Communications Materials, vol. 6, p. 11.
DOI: [10.1038/s43246-024-00707-w](https://doi.org/10.1038/s43246-024-00707-w)
7. **Gkanias, E.**, Webb, B. 2025. *Spatiotemporal computations in the insect celestial compass*. Nature Communications, vol. 16, p. 2832.
DOI: [10.1038/s41467-025-57937-w](https://doi.org/10.1038/s41467-025-57937-w)

Conference publications

8. Stouraitis, T.*, **Gkanias, E.***, Hemmi, J. M., Webb, B. 2017. *Predator evasion by a Robocrab*. Proceedings in 6th International Conference on Biomimetic and Biohybrid Systems, Living Machines, vol. 10384, p. 428-439.
DOI: [10.1007/978-3-319-63537-8_36](https://doi.org/10.1007/978-3-319-63537-8_36)
* co-first authors

In preparation

9. Chitic, I. R., **Gkanias, E.**, Ságodi, Á., Webb, B., Chicca, E. 2026. Attractor working memory as a potential basis for insect path integration. Submitted to *Biological Cybernetics*.
10. **Gkanias, E.**, Kolyfetis, G., Dacke, M., Foster, J., Webb, B. 2026. The celestial compass of insects needs a single ray of skylight to find the sun.
11. Kolyfetis, G., **Gkanias, E.**, Aliyam, A., Jie, V., Baird, E., Galizia, G., Webb, B., Foster, J. 2026. From eye anatomy to navigation: a biologically accurate model of bees' polarisation vision.

List of conference presentations

- 2017 **The Living Machines** [oral]
Title: *Predator evasion by a Robocrab.*
Location: Stanford University, CA, United States.
- 2018 **The Maggot Meeting** [poster]
Title: *Imitating the Drosophila larval learning behaviour on a robot.*
Location: University of Edinburgh, United Kingdom.
- 2019 **International Conference on Invertebrate Vision (ICIV)** [poster]
Title: *Robustness of a model of the insects' celestial compass in realistic conditions.*
Location: Backäskog Slott, Sweden.
- International Navigation Conference (INC)** [oral]
Title: *From skylight input to behavioural output: a computational model of the insect polarised light compass.*
Location: Edinburgh International Conference Centre, United Kingdom.
- 2021 **Mushroom Body Meeting** [oral]
Title: *How flies acquire, forget and assimilate memories: computational perspective.*
Location: online
- Bernstein Conference** [poster]
Title: *The incentive circuit of the fruit fly brain: a computational perspective.*
Location: online
- 2022 **Computational and Systems Neuroscience (Cosyne) Conference** [poster]
Title: *An anatomically accurate circuit for short- and long-term motivational learning in fruit flies.*
Location: Lisbon, Portugal.
- International Congress of Neuroethology (ICN)** [poster]
Title: *How could the mushroom body and central complex combine for visual homing in insects?*
Location: Lisbon, Portugal.
- Structure and Function of the Insect Central Complex** [poster]
Title: *How can the fan-shaped body integrate differential familiarity for route following in desert ants?*
Location: Janelia Research Campus, VA, United States.
- 2023 **Neural Circuits and Behaviour of Drosophila** [oral]
Title: *Memory dynamics in Drosophila's mushroom body: a computational view.*
Location: Orthodox Academy of Crete, Greece.
- International Conference on Invertebrate Vision (ICIV)** [oral]
Title: *Multimodal skylight information improves the estimation of the celestial compass: insights from a hardware implementation.*
Location: Backäskog Slott, Sweden.
- 2024 **International Congress of Neuroethology (ICN)** [poster]
Title: *Time compensation in the celestial compass of insects.*
Location: Freie Universität Berlin, Germany.
- 2025 **Electrical and Computer Engineering Student Conference (ECESC)** [keynote/invited]
Title: *The insects' alternative to GPS.*
Location: Aristotle University of Thessaloniki, Greece.
- Bioinspired Sensing Computing and Control with International Teams (BISCCITs)** [poster]
Title: *The time-compensated celestial compass of insects.*
Location: University of Colorado–Boulder, CO, USA.
- The Future of Bee and Fly Neuroethology, Modelling and Robotics** [poster]
Title: *The time-compensated celestial compass of insects.*
Location: Harnack Haus Berlin, Germany.